California State University, Fresno
Lyles College of Engineering
Electrical and Computer Engineering Department

Technical Report

Final Project

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1. Introduction

This project involves the design and simulation of a two-pole low-pass Butterworth filter using the internal equivalent circuit of the 741 operational amplifier. The goal is to achieve a flat passband response with a cutoff frequency of 25 kHz, using calculated RC values based on standard Butterworth design principles. The filter is implemented using both an ideal op-amp model and a transistor-level model of the 741 to compare practical effects. Key aspects of the op-amp's internal structure, such as the differential input stage, gain stage with Miller compensation, and class-AB output stage, are analyzed to understand their impact on filter performance.

2. Circuit Diagram and Explanation

Figure 2.1(a) shows the schematic of the two-pole low-pass Butterworth filter. This circuit uses an ideal op-amp in combination with passive components—two equal resistors R, and two capacitors with chosen values: $C_3 = 1.414C$ and $C_4 = 0.707C$.

These capacitor ratios are derived from the Butterworth condition to ensure a maximally flat magnitude response in the passband. The cutoff frequency of the filter is determined by the formula $f_{3dB} = \frac{1}{2\pi RC}$, and the magnitude response shown in Figure 2.1(b) confirms that the filter attenuates frequencies beyond the cutoff at a rate of -40 dB/decade, characteristic of a second-order system.

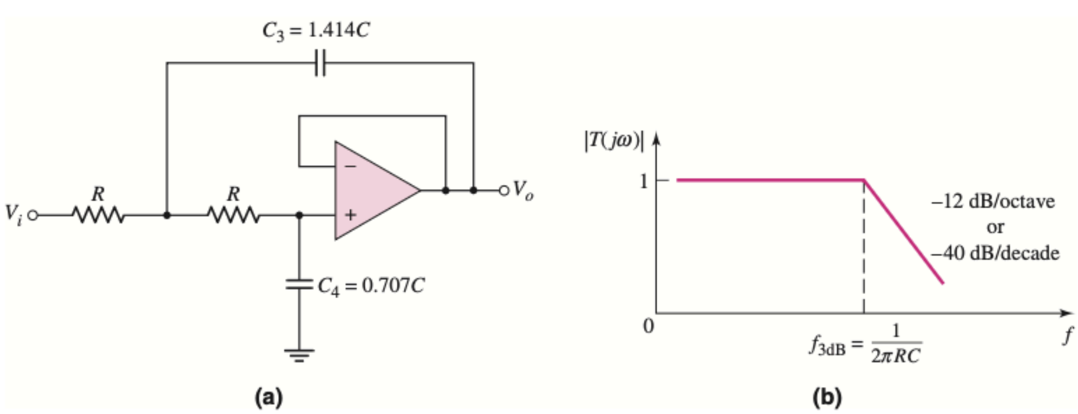

Figure 2.1: Two Pole Low Pass Butterworth Filter

To realize this filter, 741 op-amp is used as the active element. Its internal structure, shown in Figure 2.2, includes multiple transistor stages that directly support filter operation. The input differential amplifier stage (Q1–Q7), along with its biasing transistors (Q8–Q12), provides high input impedance and strong differential amplification. Transistors Q1 and Q2 act as emitter followers, feeding the common-base amplifier formed by Q3 and Q4, which provides voltage gain. The active loads (Q5–Q7 and associated resistors) allow for a single-ended output taken from the collectors of Q4 and Q6. This stage also includes provisions for offset nulling, ensuring that the output remains at zero when the differential input is zero.The gain stage, composed of transistors Q16 and Q17, receives the signal from the input stage and provides the main voltage amplification. Transistor Q16 operates as an emitter follower to prevent loading effects, while Q17 operates in a common-emitter configuration. The most critical component here is the Miller compensation capacitor C_1 connected between the output and input of this stage. This capacitor introduces a dominant pole in the open-loop gain response, stabilizing the op-amp and contributing to the low-pass behavior of the system.

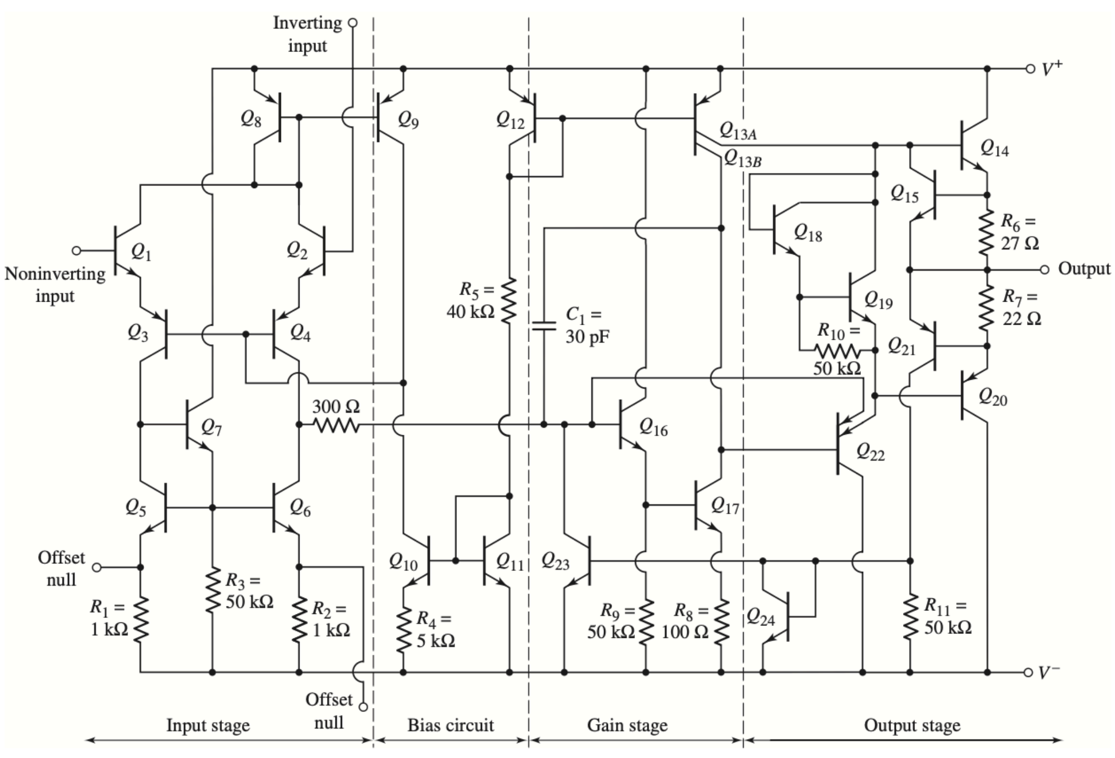

Figure 13.3 Equivalent circuit, 741 op-amp

Figure 2.2: Equivalent Circuit 741 Op Amp

Lastly, the output stage (Q14–Q22) is a class-AB complementary emitter-follower design that offers low output impedance and current gain to drive external loads. Transistors Q22 and Q13A help isolate the gain stage from the output stage, preventing loading. Additional transistors, such as Q15 and Q21, provide short-circuit protection.

3. Design Computations

We are asked to design a two-pole low-pass Butterworth filter with a bandwidth of 25kHz:

1. Use Cutoff Frequency Formula to Solve for RC

$$f_{3dB} = \frac{1}{2\pi RC} \Rightarrow RC = \frac{1}{2\pi f_{3dB}} = \frac{1}{2\pi\times(25\times10^{3}Hz)} = 6.37 \times 10^{-6}$$

2. Use Butterworth Capacitor Ratios: $C_3 = 1.414C$ and $C_4 = 0.707C$

   Given largest capacitor to be used $50pF$, Solve for C:

   $$C = \frac{C_3}{1.414} = \frac{50pF}{1.414} \approx 35.36pF$$

   Now Calculate C_4:

   $$C_4 = 0.707C = 0.707 \times 35.36pF \approx 25pF$$

3. Solve For R

   $$R = \frac{RC}{C} = \frac{6.37\times10^{-6}}{35.36\times10^{-12}} = 180k\Omega$$

4. Final Parameters

   So the final Parameters for the Circuit are:
   - $R = 180k\Omega$
   - $C_3 = 50pF$
   - $C_4 = 25pF$

4. Computer Simulation and Results

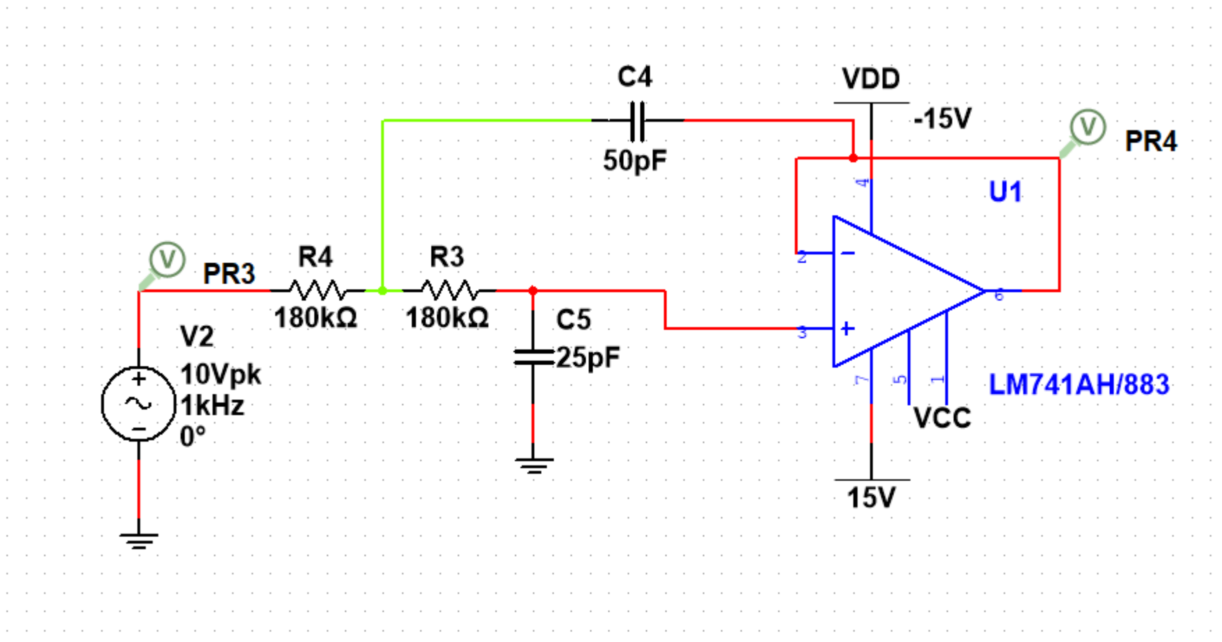

Figure 4.1 Two Pole Low Pass Butterworth Filter

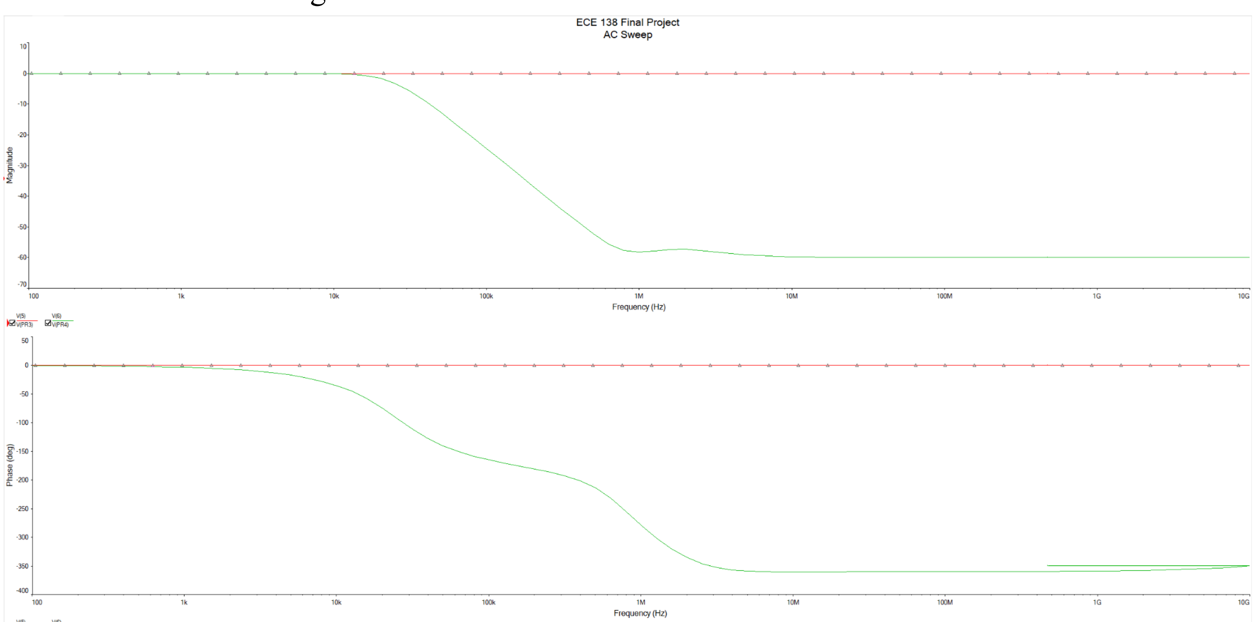

Figure 4.2 - Bode Plot of Two Pole Low Pass Butterworth Filter

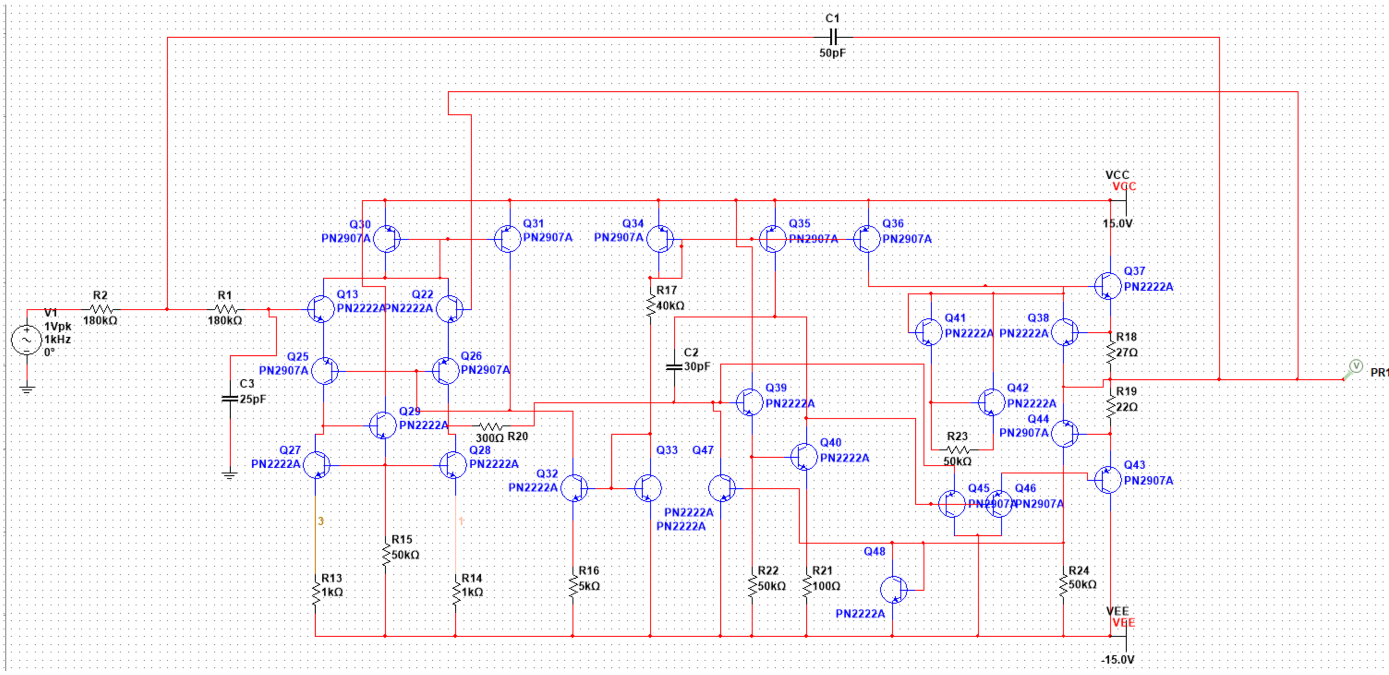

Figure 4.3 - Two Pole Low Pass Butterworth Filter using Equivalent Circuit of 741 Op Amp

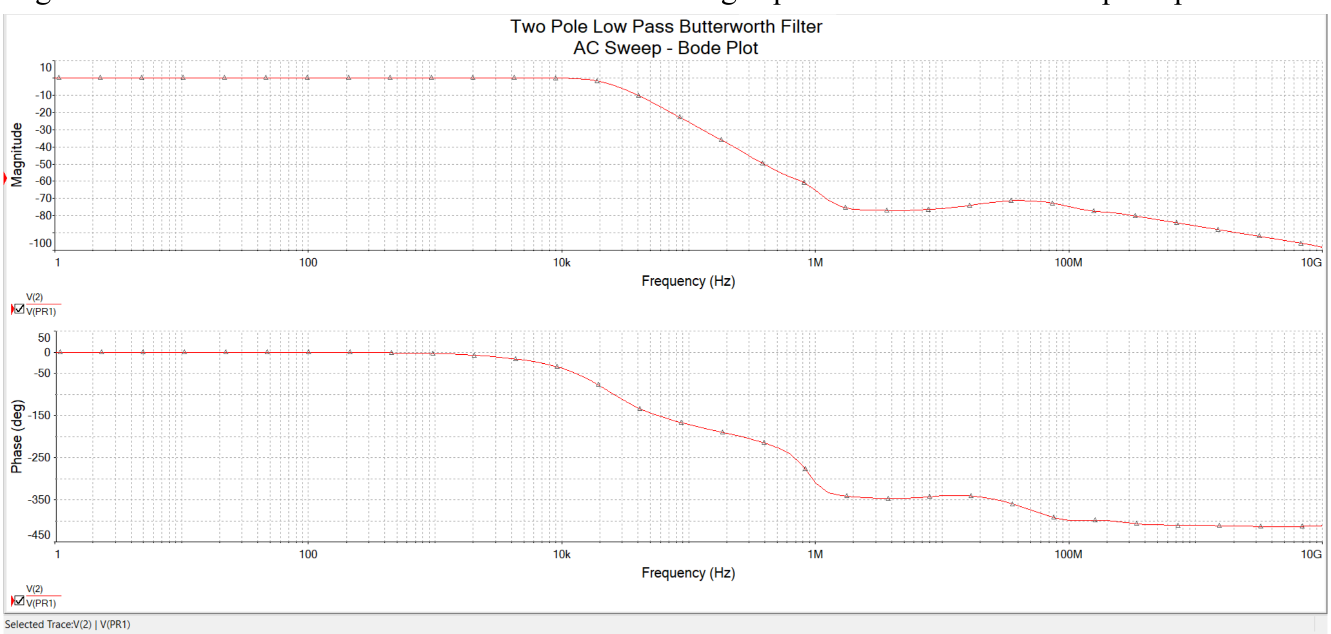

Figure 4.4 - Bode Plot of Two Pole Low Pass Butterworth Filter using Equivalent Circuit of 741 Op Amp

5. Conclusion

The two-pole Butterworth filter was successfully designed and simulated to meet a 25 kHz cutoff frequency requirement. Both the ideal op-amp and the internal transistor-level model of the 741 demonstrated the expected low-pass filter characteristics, with simulation results confirming a -40 dB/decade roll-off. The internal stages of the 741 op-amp were shown to play a key role in filter performance, particularly the Miller compensation capacitor in the gain stage.

7. References

1. Neamen, Donald A. *Microelectronics : Circuit Analysis and Design*. New York, Mcgraw-Hill, 2010, pp..